

Week 11 Tutorial

Earnings dilution, incentive compensation, and project selection

Zirui Song
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Presentation

- The presentations will all be held in **Week 12**.
- Each group will have a maximum of **8 minutes**.
- The presentation order will be determined by **drawing lots** for the groups.
- All group members must present; no one is exempt.
- The Maximum number of slides is **15**
- The structure and the contents of the slides and any other aspects of the slides (colour etc) are **your group's decision**.
- Email me your complete slides in PPT before the presentation class.
- If your group needs to present in Week 11 instead, please let me know in advance (by Sunday, May 11).

Agenda

- 7-2
- 7-5

Review 1: Economic value added (EVA)

1. Why managers care about earnings

- An investment project that reduces the firm's earnings is referred to as **earnings dilutive**. Likewise, an investment that increases near-term earnings is referred to as **earning accretive**.
- Managers care about reported earnings because that they are often paid to care about earnings. Companies frequently tie short-term bonuses either directly or indirectly to the firm's operating performance, as reflected in its earnings.
- Besides, earnings may affect the CEO's ability to operate without too much scrutiny from the board of directors, and perhaps even to keep his or her job.

Review 1: Economic value added (EVA)

2. Equity-cost problem

- It refers to a mismatch between accounting-based performance and true economic value creation.
- It occurs when managers evaluate projects based on their impact on reported earnings, without accounting for the **opportunity cost** of equity capital—leading to the acceptance of value-destroying projects that appear profitable under accounting metrics.
- *Bad projects with good earnings*
- Example: Beck Electronics

Review 1: Beck Electronics

Table 7-1 Example 1: Bad Project with Good Earnings for Beck Electronics

Panel (a) Project Characteristics and Assumptions

Investment outlay	\$6,000,000	One-time expenditure
Project life	Infinite	Perpetual life asset
CAPEX = depreciation expense (for project)	—	
Debt financing	2,000,000	Perpetual debt (never matures)
Equity (from retained earnings)	4,000,000	Raised using excess cash
Additional EBIT per year	500,000	Per year (level perpetuity)
Tax rate	20.0%	
Treasury bill yield	4.0%	
Borrowing rate (before tax)	5.0%	
Cost of equity	10.0%	
Cost of capital (WACC)	8.0%	WACC is the same for the firm and the project

Review 1: Beck Electronics

Panel (b) Pro Forma Income Statements	Pre-Project Firm	Project	Firm + Project
EBIT	\$4,000,000	\$ 500,000	\$4,500,000
Less: interest expense	(500,000)	(100,000)	(600,000)
Plus: interest income (equity financing)	160,000	0	0
EBT	\$3,660,000	\$ 400,000	\$3,900,000
Less: taxes	(732,000)	(80,000)	(780,000)
Net income	\$2,928,000	\$ 320,000	\$3,120,000
Earnings per share	\$ 1.464	\$ 0.160	\$ 1.560

$= 4,000,000 \times 0.04$

Panel (c) Cash Flow Analysis	Pre-Project Firm	Project	Firm + Project
EBIT	\$4,000,000	\$ 500,000	\$4,500,000
Less: taxes	(800,000)	(100,000)	(900,000)
NOPAT	\$3,200,000	\$ 400,000	\$3,600,000
Plus: depreciation	2,400,000	0	2,400,000

Note: EPS increases if the project is accepted!

Review 1: Beck Electronics

Less: CAPEX	(2,400,000)	(0)	(2,400,000)
FFCF	\$ 3,200,000	\$ 400,000	\$ 3,600,000

Panel (d) Firm and Project Valuation Analysis

	Project	Firm + Project
Value of pre-project firm		$\frac{3,200,000}{0.08} = \$40,000,000$
Less: investment outlay	$\frac{400,000}{0.08} = \$6,000,000$	
Plus: value of project cash flows	5,000,000	
Value of firm plus project		39,000,000
Net present value		<u>$\\$(1,000,000)$</u>

Legend:

EBIT = earnings before interest and taxes

EBT = earnings before taxes

NOPAT = net operating profit after taxes

CAPEX = capital expenditures

FFCF = firm free cash flow

Note: The NPV of the project is negative!

Review 1: Beck Electronics

$$\begin{aligned} \text{Change in Net Income} &= \underbrace{\left[\left(\frac{\text{Project Operating Income}}{\text{Operating Income}} \right) (1 - \text{Tax Rate}) \right]}_{\text{NOPAT}} \\ &\quad - \underbrace{\left[\left(\frac{\text{Interest Expense on New Debt}}{\text{Interest Expense on New Debt}} \right) (1 - \text{Tax Rate}) + \left(\frac{\text{Lost Interest Income on Cash Used to Fund the Equity}}{\text{Lost Interest Income on Cash Used to Fund the Equity}} \right) (1 - \text{Tax Rate}) \right]}_{\text{Capital Cost}} \\ &= \text{NOPAT} - \text{Capital Cost} \end{aligned} \tag{7.1b}$$

Understate the true opportunity cost

Review 1: Beck Electronics

- Stern Stewart & Co. define economic profit (EVA) as follows:

$$\text{EVA}_t^{\text{®}} = \frac{\text{Net Operating Profit after Taxes (NOPAT)}_t}{\text{Capital Charge}_t} \quad (7.2)$$

where

$$\text{NOPAT}_t = \left(\frac{\text{Earnings Before Interest and Taxes (EBIT)}_t}{\text{EBIT}_t} \right) \times \left(1 - \text{Corporate Tax Rate} \right)$$

and

$$\text{Capital Charge}_t = \frac{\text{Invested Capital}_{t-1}}{\text{Invested Capital}_{t-1}} \times \text{Weighted Average Cost of Capital (WACC)}$$

Review 1: Beck Electronics

Table 7-6 Using Economic Profit to Solve the Equity-Cost Problem for Beck Electronics Inc.

$$= 6,000,000 \times 0.08$$

	Pre-Project Firm	Project	Firm + Project
EBIT	\$ 4,000,000	\$ 500,000	\$ 4,500,000
Less: taxes	(800,000)	(100,000)	(900,000)
NOPAT	\$ 3,200,000	\$ 400,000	\$ 3,600,000
Less: capital charge	(2,400,000)	(480,000)	(2,880,000)
Economic profit	\$ 800,000	\$ (80,000)	\$ 720,000

EBIT = earnings before interest and taxes; NOPAT = net operating profit after taxes; capital charge = invested capital times the weighted average cost of capital; equity capital charge = equity capital times the cost of equity capital; economic profit = NOPAT less the capital charge.

Review 1: Economic value added (EVA)

3. Back-loaded problem

- Back-loaded projects generate most of their cash flows in later years, so even if they have a positive NPV, their early-year earnings may appear weak or even negative, making them dilutive to EPS and possibly misleading for short-term performance evaluation.
- This fact poses a very serious dilemma for the financial manager who would like to select positive NPV investment but is evaluated and paid based on the firm's year-to-year performance.
- *Good projects with back-loaded earnings*
- Example: Dowser Chemical

Review 1: Dowser Chemical

Table 7-4 Analysis of Project Cash Flow and NPV for Dowser Chemical (\$ Thousands)

	Years			
	1	2	3	4
Sales	\$ 30,000	\$ 36,000	\$ 43,200	\$ 51,840
Net operating income	0	1,600	3,520	5,824
Less cash tax payments	0	(480)	(1,056)	(1,747)
Net operating profits after taxes (NOPAT)	\$ 0	\$ 1,120	\$ 2,464	\$ 4,077
Plus: depreciation expense	3,000	3,200	3,440	3,728
Less: investments:				
In net working capital	(500)	(600)	(720)	(864)
In new capital (CAPEX)	(5,000)	(5,600)	(6,320)	(7,184)
Total net investment for the period	\$ (5,500)	\$ (6,200)	\$ (7,040)	\$ (8,048)
Free cash flow (FCF)	\$ (2,500)	\$ (1,880)	\$ (1,136)	\$ (243)

Project Valuation

Present value of invested capital, year 10	\$ 6,859.73
Terminal value of free cash flows (12% cost of capital)	31,359.13
Present value of project cash flows	\$ 38,218.85
Less: initial invested capital in year 0	(32,500.00)
Net present value	<u>\$ 5,718.86</u>

	Years					
	5	6	7	8	9	10
Sales	\$ 62,208	\$ 74,650	\$ 89,580	\$ 107,495	\$ 128,995	\$ 154,793
Net operating income	8,589	11,907	15,888	20,665	26,399	33,278
Less cash tax payments	(2,577)	(3,572)	(4,766)	(6,200)	(7,920)	(9,983)
Net operating profits after taxes (NOPAT)	\$ 6,012	\$ 8,335	\$ 11,122	\$ 14,466	\$ 18,479	\$ 23,295
Plus: depreciation expense	4,074	4,488	4,986	5,583	6,300	7,160
Less: investments:						
In net working capital	(1,037)	(1,244)	(1,493)	(1,792)	(2,150)	(2,580)
In new capital (CAPEX)	(8,221)	(9,465)	(10,958)	(12,750)	(14,899)	(17,479)
Total net investment for the period	\$ (9,258)	\$ (10,709)	\$ (12,451)	\$ (14,541)	\$ (17,049)	\$ (20,059)
Free cash flow (FCF)	\$ 828	\$ 2,114	\$ 3,657	\$ 5,508	\$ 7,729	\$ 10,395

Review 1: Dowser Chemical

Table 7-5 Analysis of Earnings Dilution Effects for Dowser Chemical (\$ Thousands Except for Per-Share Information)

Analysis of Earnings Dilution Effects	1	2	3	4	5	6	7	8	9	10
Project net income	\$ (758)	\$ 303	\$ 1,577	\$ 3,106	\$ 4,941	\$ 7,142	\$ 9,784	\$ 12,954	\$ 16,758	\$ 21,323
Firm net income without the project	53,750	57,781	62,115	66,773	71,781	77,165	82,952	89,174	95,862	103,052
Total firm net income with the project	\$ 52,992	\$ 58,085	\$ 63,692	\$ 69,880	\$ 76,722	\$ 84,307	\$ 92,736	\$ 102,128	\$ 112,620	\$ 124,375
Total shares outstanding with the project	10,688.33	10,736.81	10,760.31	10,760.31	10,760.31	10,760.31	10,760.31	10,760.31	10,760.31	10,760.31
Combined firm and project earnings per share	\$ 4.96	\$ 5.41	\$ 5.92	\$ 6.49	\$ 7.13	\$ 7.84	\$ 8.62	\$ 9.49	\$ 10.47	\$ 11.56
Total shares outstanding without the project	10,000	10,000	10,000	10,000	10,000	10,000	10,000	10,000	10,000	10,000
Firm earnings per share without the project	\$ 5.38	\$ 5.78	\$ 6.21	\$ 6.68	\$ 7.18	\$ 7.72	\$ 8.30	\$ 8.92	\$ 9.59	\$ 10.31
Earnings (dilution)/accretion per share	\$ (0.42)	\$ (0.37)	\$ (0.29)	\$ (0.18)	\$ (0.05)	\$ 0.12	\$ 0.32	\$ 0.57	\$ 0.88	\$ 1.25

Review 1: Dowser Chemical

Table 7-7 Using Economic Profit (or EVA[®]) to Evaluate the Dowser Chemical Example Introduced in Table 7-3

Panel (a) Calculating Economic Profit (\$000)										
Method 1	Years				Years					
	1	2	3	4	5	6	7	8	9	10
Sales	\$ 30,000	\$ 36,000	\$ 43,200	\$ 51,840	\$ 62,208	\$ 74,650	\$ 89,580	\$ 107,495	\$ 128,995	\$ 154,793
Operating income	0	1,600	3,520	5,824	8,589	11,907	15,888	20,665	26,399	33,278
Less: cash tax payments	0	(480)	(1,056)	(1,747)	(2,577)	(3,572)	(4,766)	(6,200)	(7,920)	(9,983)
Net operating profits after taxes (NOPAT)	\$ 0	\$ 1,120	\$ 2,464	\$ 4,077	\$ 6,012	\$ 8,335	\$ 11,122	\$ 14,466	\$ 18,479	\$ 23,295
Less: capital cost = invested capital $\times k_{WACC}$	(3,900)	(4,200)	(4,560)	(4,992)	(5,510)	(6,132)	(6,879)	(7,775)	(8,850)	(10,140)
Economic profit = NOPAT – capital cost	\$ (3,900)	\$ (3,080)	\$ (2,096)	\$ (915)	\$ 502	\$ 2,202	\$ 4,243	\$ 6,691	\$ 9,629	\$ 13,155
Method 2										
Return on invested capital (ROIC)	0.00%	3.20%	6.48%	9.80%	13.09%	16.31%	19.40%	22.33%	25.06%	27.57%
Cost of capital (k_{WACC})	12%	12%	12%	12%	12%	12%	12%	12%	12%	12%
Invested capital ($t - 1$)	\$ 32,500	\$ 35,000	\$ 38,000	\$ 41,600	\$ 45,920	\$ 51,104	\$ 57,325	\$ 64,790	\$ 73,748	\$ 84,497
Economic profit = (ROIC – k_{WACC}) $\times$ invested capital ($t - 1$)	\$ (3,900)	\$ (3,080)	\$ (2,096)	\$ (915)	\$ 502	\$ 2,202	\$ 4,243	\$ 6,691	\$ 9,629	\$ 13,155
Panel (b) Market Value Added (MVA) and Project NPV (\$000)										
MVA = present value of economic profits	\$ 5,719									
Plus: invested capital	32,500									
Project value	\$38,219									
NPV = project value – invested capital	\$ 5,719									

Even though EVA is better than accounting profit, it still punishes back-loaded projects.

Review 1: A front-loaded project example

Table 7-8 Using Economic Profit (or EVA[®]) to Evaluate an Investment with Front-Loaded Earnings

Panel (a) Calculating Economic Profit (\$000)

Method 1	Years			
	1	2	3	4
Sales	\$ 80,000	\$ 76,000	\$ 72,200	\$ 68,590
Operating income	12,632	11,600	10,620	9,689
Less: cash tax payments	(3,789)	(3,480)	(3,186)	(2,907)
Net operating profits after taxes (NOPAT)	\$ 8,842	\$ 8,120	\$ 7,434	\$ 6,782
Less: capital charge = invested capital $\times k_{WACC}$	(7,453)	(7,200)	(6,960)	(6,732)
Economic profit	\$ 1,389	\$ 920	\$ 474	\$ 50

Method 2				
Return on invested capital (ROIC)	14.24%	13.53%	12.82%	12.09%
Cost of capital (k_{WACC})	12%	12%	12%	12%
Invested capital	\$ 62,105	\$ 60,000	\$ 58,000	\$ 56,100
Economic profit	\$ 1,389	\$ 920	\$ 474	\$ 50

Panel (b) Market Value Added (MVA) and Project NPV (\$000)

MVA = present value of economic profits	\$ (617)
Plus: invested capital	62,105
Project value	\$ 61,488
NPV = project value – invested capital	\$ (617)

Years					
5	6	7	8	9	10
\$ 65,161	\$ 61,902	\$ 58,807	\$ 55,867	\$ 53,074	\$ 50,420
8,805	7,964	7,166	6,408	5,687	5,003
(2,641)	(2,389)	(2,150)	(1,922)	(1,706)	(1,501)
\$ 6,163	\$ 5,575	\$ 5,016	\$ 4,485	\$ 3,981	\$ 3,502
(6,515)	(6,310)	(6,114)	(5,928)	(5,752)	(5,584)
\$ (352)	\$ (735)	\$ (1,098)	\$ (1,443)	\$ (1,771)	\$ (2,082)

11.35%	10.60%	9.85%	9.08%	8.31%	7.53%
12%	12%	12%	12%	12%	12%
\$ 54,295	\$ 52,580	\$ 50,951	\$ 49,404	\$ 47,933	\$ 46,537
\$ (352)	\$ (735)	\$ (1,098)	\$ (1,443)	\$ (1,771)	\$ (2,082)

Review 1: A front-loaded project example

- This mismatch between standard accounting methods for invested capital in depreciable assets—typically based on straight-line or accelerated depreciation—and the actual economic pattern of value loss leads to distorted measures of both earnings and economic profit.
- To correct this distortion and align performance measurement with the true timing of value creation, a **revised EVA** approach adjusts for **economic depreciation**, ensuring more accurate evaluation of investment decisions over the project lifecycle.

Review 1: Revised EVA

- Standard EVA formula (based on accounting depreciation)

$$EVA_t^{\text{®}} = \frac{\text{Net Operating Profit after Taxes (NOPAT)}_t}{\text{Capital Charge}_t} \quad (7.2)$$

- Economic Depreciation formula

$$\text{Economic Depreciation}_t = PV_t(\text{Future FCFs}) - PV_{t-1}(\text{Future FCFs})$$

- Capital Charge formula

$$\text{Revised Capital Charge}_t = PV_{t-1}(\text{Future FCFs}) \times WACC$$

- Revised EVA formula

$$\text{Revised EVA}_t = \frac{FCF_t + \text{Economic Depreciation}_t}{\text{Revised NOPAT}} - \text{Revised Capital Charge}_t$$

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Table 7-9 Revising Economic Profit to Reflect Economic Depreciation

Panel (a) Back-Loaded Earnings Example (\$ Thousands)											
	0	1	2	3	4	5	6	7	8	9	10
Project free cash flows	\$ (32,500.00)	\$ (2,500.00)	\$ (1,880.00)	\$ (1,136.00)	\$ (243.20)	\$ 828.16	\$ 2,113.79	\$ 3,656.55	\$ 5,507.86	\$ 7,729.43	\$107,792.02
Internal rate of return on the investment	13.73%										
Present value of future cash flows (estimated invested capital)	\$ 32,500.00	\$ 39,462.08	\$ 46,760.02	\$ 54,315.93	\$ 62,016.42	\$ 69,702.79	\$ 77,158.83	\$ 84,095.79	\$ 90,133.84	\$ 94,779.32	\$ 0.00
Economic depreciation = change in present value of future cash flows	1	6,962.08	7,297.94	7,555.91	7,700.49	7,686.37	7,456.04	6,936.96	6,038.05	4,645.48	(94,779.32)
Revised NOPAT = project FCF + economic depreciation	2	\$ 4,462.08	\$ 5,417.94	\$ 6,419.91	\$ 7,457.29	\$ 8,514.53	\$ 9,569.83	\$ 10,593.51	\$ 11,545.91	\$ 12,374.91	\$ 13,012.71
Revised capital charge = revised invested capital($t - 1$) × cost of capital	3	(3,900.00)	(4,735.45)	(5,611.20)	(6,517.91)	(7,441.97)	(8,364.34)	(9,259.06)	(10,091.49)	(10,816.06)	(11,373.52)
Revised estimate of economic profit		\$ 562.08	\$ 682.49	\$ 808.71	\$ 939.38	\$ 1,072.56	\$ 1,205.50	\$ 1,334.45	\$ 1,454.42	\$ 1,558.85	\$ 1,639.19
NPV = MVA = present value of all future economic profits (revised)	\$ 5,718.85										

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Panel (b) Front-Loaded Earnings Example (\$000)											
	0	1	2	3	4	5	6	7	8	9	10
Project free cash flows	\$ (62,105.26)	\$ 10,947.37	\$ 10,120.00	\$ 9,334.00	\$ 8,587.30	\$ 7,877.94	\$ 7,204.04	\$ 6,563.84	\$ 5,955.64	\$ 5,377.86	\$ 50,038.95
Internal rate of return on the investment	11.79%										
Present value of future cash flows	\$ 62,105.26	\$ 58,481.68	\$ 55,258.16	\$ 52,440.50	\$ 50,037.27	\$ 48,060.00	\$ 46,523.46	\$ 45,445.92	\$ 44,849.51	\$ 44,760.54	\$ 0.00
Economic depreciation		(3,623.58)	(3,223.52)	(2,817.66)	(2,403.23)	(1,977.27)	(1,536.54)	(1,077.54)	(596.42)	(88.97)	(44,760.54)
Revised NOPAT = project FCF + economic depreciation		\$ 7,323.79	\$ 6,896.48	\$ 6,516.34	\$ 6,184.07	\$ 5,900.67	\$ 5,667.50	\$ 5,486.30	\$ 5,359.23	\$ 5,288.90	\$ 5,278.41
Less: revised capital charge		\$ (7,452.63)	\$ (7,017.80)	\$ (6,630.98)	\$ (6,292.86)	(6,004.47)	(5,767.20)	(5,582.82)	(5,453.51)	(5,381.94)	(5,371.26)
Revised estimate of economic profit		\$ (128.84)	\$ (121.33)	\$ (114.64)	\$ (108.79)	\$ (103.81)	\$ (99.70)	\$ (96.52)	\$ (94.28)	\$ (93.04)	\$ (92.86)
MVA = present value of all future economic profits (revised)	\$ (617.10)										

Table 7-10 Summary of the Managerial Horizon Problem:
Solving the Measurement Problem

Distribution of Project Earnings/Cash Flows over Time	Source of Financing	Problem	Solution
I. Stable (i.e., neither front- nor back-loaded)	Internal equity	Economic profit assigns the proper equity cost to the project, whereas EPS assigns a cost for equity capital equal to the forgone interest income earned by the firm on excess cash.	Evaluate periodic performance using economic profit.
	External equity (sale of common stock)	If the cost of equity is not equal to the E/P ratio, then the EPS dilution (accretion) criterion is not consistent with $NPV < 0 (> 0)$.	Evaluate periodic performance using economic profit.
	Debt financing	Economic profit assigns the proper cost of equity. However, projects financed with debt will be accretive as long as they earn more than the after-tax cost of debt.	Evaluate periodic performance using economic profit.

II. Back-loaded	Any	The annual economic profit in the early years of the project's life will be negative, even for positive NPV projects.	Revise economic profit metric to reflect economic depreciation. Capitalize R&D and advertising expenses and amortize over the project's life. Use a bonus bank compensation plan to extend management's horizon.
III. Front-loaded	Any	The annual economic profit in the early years of the project's life can be positive, even for negative NPV projects.	Revise economic profit metric to reflect economic depreciation. Capitalize R&D and advertising expenses and amortize over the project's life. Use a bonus bank compensation plan to extend management's horizon.

Problem 1

7-2 ANALYZING A PROJECT WITH BACK-LOADED EARNINGS Hospital Services Inc. provides health care services primarily in the western part of the United States. The firm operates psychiatric hospitals that provide mental health care services using inpatient, partial hospitalization, and outpatient settings. In spring 2015, the firm is considering an investment in a new patient-monitoring system that costs \$6 million per hospital to install. The new system is expected to contribute to the firm's earnings before interest, taxes, depreciation, and amortization (EBITDA) via annual savings of \$2.4 million in years 1 and 2, plus \$4.25 million in year 3.

The firm's CFO is interested in investing in the new system but is concerned that the savings from the system are such that the immediate impact of the project may be to dilute the firm's earnings. The firm has just moved to an economic profit-based bonus system, and the CFO fears that the project may also make the individual economic profits of the hospitals look bad—a development that would generate resistance from the various hospital managers if they saw their bonuses being decreased.

- a. Assume that the cost of capital for the project is 15%, the firm faces a 30% marginal tax rate, it uses straight-line depreciation over a three-year life for the new investment, and it has a zero salvage value. Calculate the project's expected NPV and IRR.

Problem 1

- b. Calculate the annual economic profits for the investment for years 1 through 3. What is the present value of the annual economic profit measures discounted using the project's cost of capital? What potential problems do you see for the project?
- c. Calculate the economic depreciation for the project, and use it to calculate a revised economic profit measure following the procedure laid out in Table 7-8. What is the present value of all the revised economic profit measures when discounted using the project's cost of capital? (*Hint:* First, revise the initial NOPAT estimate from your answer to Exercise 7-2a by subtracting the economic depreciation estimate from project free cash flow calculated in Exercise 7-2a. Next, calculate the capital charge for each year based on invested capital less economic depreciation.)
- d. Using your analysis from Exercise 7-2b and c, calculate the return on invested capital (ROIC) for years 1 through 3 as the ratio of NOPAT for year t to invested capital for year $t - 1$. Compare the two sets of calculations and discuss how the use of economic depreciation affects the ROIC estimate for the project.

Problem 1

Answer

- a. Given value: *Initial investment*₂₀₁₅ = 6,000,000; *EBITDA*₂₀₁₆ = *EBITDA*₂₀₁₇ = 2,400,000; *EBITDA*₂₀₁₈ = 4,250,000; *Cost of capital* = 15%; *Tax rate* = 30%; *Straight – line depreciation (three – year life and a zero salvage value)*
- $EBIT_{2016} = EBITDA_{2016} - Depreciation = 2,400,000 - \frac{6,000,000}{3} = 400,000$; $NOPAT_{2016} = 400,000 \times (1 - 30\%) = 280,000$; $FCF_{2016} = NOPAT_{2016} + Depreciation = 280,000 + 2,000,000 = 2,280,000$
 - Similarly, the calculation of FCF also reveal that: $FCF_{2017} = 2,280,000$; $FCF_{2018} = 1,575,000$

Problem 1

- $NPV = \frac{2,280,000}{(1+15\%)^1} + \frac{2,280,000}{(1+15\%)^2} + \frac{4,250,000}{(1+15\%)^3} - 6,000,000 = 57,236.79 > 0$
- $NPV = 0 = \frac{2,280,000}{(1+r\%)^1} + \frac{2,280,000}{(1+r\%)^2} + \frac{4,250,000}{(1+r\%)^3} - 6,000,000; IRR(r) = 15.53\% > 15\%$

b. EVA

$$EVA_t^{\text{®}} = \frac{\text{Net Operating Profit after Taxes (NOPAT)}_t}{\text{Capital Charge}_t} \quad (7.2)$$

where

$$NOPAT_t = \left(\frac{\text{Earnings Before Interest and Taxes (EBIT)}_t}{\text{EBIT}_t} \right) \times \left(1 - \frac{\text{Corporate Tax Rate}}{\text{Tax Rate}} \right)$$

and

$$\text{Capital Charge}_t = \frac{\text{Invested Capital}_{t-1}}{\text{Capital}_{t-1}} \times \frac{\text{Weighted Average Cost of Capital (WACC)}}{\text{Capital (WACC)}}$$

Problem 1

- $NOPAT_{2016} = 280,000$; $Capital\ Charge_{2016} = 6,000,000 \times 15\% = 900,000$; $EVA_{2016} = 280,000 - 900,000 = -620,000$
- $NOPAT_{2017} = 280,000$; $Capital\ Charge_{2017} = (6,000,000 - 2,000,000) \times 15\% = 600,000$; $EVA_{2017} = 280,000 - 600,000 = -320,000$
- $NOPAT_{2018} = 1,575,000$; $Capital\ Charge_{2018} = (4,000,000 - 2,000,000) \times 15\% = 300,000$; $EVA_{2018} = 1,575,000 - 300,000 = 1,275,000$
- $MVA = \frac{-620,000}{(1+0.15)^1} + \frac{-620,000}{(1+0.15)^2} + \frac{1,275,000}{(1+0.15)^3} = 57,236.79$
- Since EVA for the project is negative in the first two years, the project may appear unattractive to managers being evaluated annually.

Problem 1

- This highlights a major drawback of using unadjusted EVA for back-loaded projects and calls for adjustments such as economic depreciation.
- c. *The value of the project's future cash flows in year 0:* $\frac{2,280,000}{(1+15.53\%)^1} + \frac{2,280,000}{(1+15.53\%)^2} + \frac{4,250,000}{(1+15.53\%)^3} = 6,000,000$; *In year 1:* $\frac{2,280,000}{(1+15.53\%)^1} + \frac{4,250,000}{(1+15.53\%)^2} = 4,651,888.26$; *In year 2:* $\frac{4,250,000}{(1+15.53\%)^1} = 3,094,394.94$
- *Economic depreciation₂₀₁₆ = 4,651,888.26 – 6,000,000 = – 1,348,111.74; Economic depreciation₂₀₁₇ = 3,094,394.94 – 4,651,888.26 = –1,557,493.32; Economic depreciation₂₀₁₈ = – 3,094,394.94*

Problem 1

- $\text{Revised NOPAT}_{2016} = \text{FCF}_{2016} + \text{Economic depreciation}_{2016} = 2,280,000 - 1,348,111.74 = 931,888.26$; $\text{Revised NOPAT}_{2017} = 2,280,000 - 1,557,493.32 = 722,506.68$; $\text{Revised NOPAT}_{2018} = 3,575,000 - 3,094,394.94 = 480,605.06$
- $\text{Revised capital charge}_{2016} = 6,000,000 \times 15\% = 900,000$; $\text{Revised capital charge}_{2017} = 4,651,888.26 \times 15\% = 697,783.24$; $\text{Revised capital charge}_{2018} = 3,094,394.94 \times 15\% = 464,159.24$
- $\text{Revised EVA}_{2016} = 931,888.26 - 900,000 = 31,888.26$; $\text{Revised EVA}_{2017} = 722,506.68 - 697,783.24 = 24,723.44$; $\text{Revised EVA}_{2018} = 480,605.06 - 464,159.24 = 16,445.81$

Problem 1

- $MVA = \frac{31,888.26}{(1+0.15)^1} + \frac{24,723.44}{(1+0.15)^2} + \frac{16,445.81}{(1+0.15)^3} = 57,236.79$
- d. $ROIC_{2016} = \frac{NOPAT_{2016}}{Invested\ capital_{2015}} = \frac{280,000}{6,000,000} = 4.67\%$; $ROIC_{2017} = 7\%$; $ROIC_{2018} = 78.75\%$
- $Revised\ ROIC_{2016} = \frac{Revised\ NOPAT_{2016}}{Revised\ invested\ capital_{2015}} = \frac{931,888.26}{6,000,000} = 15.53\%$; $Revised\ ROIC_{2017} = 15.53\%$; $Revised\ ROIC_{2018} = 15.53\%$
- Using the traditional EVA analysis (Exercise 7-2b), the ROIC for years 1 through 3 is highly variable: 4.67%, 7.00%, and 78.75%, respectively. These fluctuations reflect the back-loaded nature of the project's earnings and the limitations of using accounting depreciation, which spreads costs evenly across years regardless of when economic value is actually created.

Problem 1

- In contrast, the revised EVA analysis (Exercise 7-2c), which incorporates economic depreciation, results in **a constant ROIC** of 15.53% across all three years. This rate exactly matches the project's internal rate of return (IRR), providing a more consistent and economically meaningful measure of performance.
- The inclusion of **economic depreciation** smooths out the distortions caused by accounting rules and better reflects the project's true value creation. It prevents the understatement of early-year performance and overstatement in later years. Overall, using economic depreciation leads to a more accurate and fair assessment of the project's profitability and return on invested capital.

Problem 2

7-5 ECONOMIC PROFIT AND NPV Steele Electronics is considering investing in a new component that requires a \$100,000 investment in new capital equipment, as well as additional net working capital. The investment is expected to provide cash flows over the next five years. The anticipated earnings and project free cash flows for the investment are found in the table below.

- a. Assuming a project cost of capital of 11.24%, calculate the project's NPV and IRR.
- b. Steele is considering the adoption of economic profit as a performance evaluation tool. Calculate the project's annual economic profit using the invested capital figures found in the table. How are your economic profit estimates related to the project's NPV?
- c. How would your assessment of the project's worth be affected if the economic profits in 2016 and 2017 were both negative? (No calculations required.)

Given

Tax Rate	30.00%					
Capital Expenditures in 2009	(\$100,000)	and none thereafter				
Weighted Average Cost of Capital	11.24%					

Project Pro Forma Income Statements

Typo: 2016-2020

	Year				
	2010	2011	2012	2013	2014
Revenues	\$ 100,000.00	\$ 105,000.00	\$ 110,250.00	\$ 115,762.50	\$ 121,550.63
Less: Cost of Goods Sold	(40,000.00)	(42,000.00)	(44,100.00)	(46,305.00)	(48,620.25)
Gross Profit	\$ 60,000.00	\$ 63,000.00	\$ 66,150.00	\$ 69,457.50	\$ 72,930.38
Less: Operating Expenses	(20,000.00)	(21,000.00)	(22,050.00)	(23,152.50)	(24,310.13)
Less: Depreciation Expense	(20,000.00)	(20,000.00)	(20,000.00)	(20,000.00)	(20,000.00)
Net Operating Income	\$ 20,000.00	\$ 22,000.00	\$ 24,100.00	\$ 26,305.00	\$ 28,620.25
Less: Interest Expense	(3,200.00)	(3,200.00)	(3,200.00)	(3,200.00)	(3,200.00)
Earnings before Taxes	\$ 16,800.00	\$ 18,800.00	\$ 20,900.00	\$ 23,105.00	\$ 25,420.25
Less: Taxes	(5,040.00)	(5,640.00)	(6,270.00)	(6,931.50)	(7,626.08)
Net Income	\$ 11,760.00	\$ 13,160.00	\$ 14,630.00	\$ 16,173.50	\$ 17,794.17

Project Free Cash Flows

	Year					
	2015	2010	2011	2012	2013	2014
Net Operating Income		\$ 20,000.00	\$ 22,000.00	\$ 24,100.00	\$ 26,305.00	\$ 28,620.25
Less: Taxes		(6,000.00)	(6,600.00)	(7,230.00)	(7,891.50)	(8,586.08)
NOPAT		\$ 14,000.00	\$ 15,400.00	\$ 16,870.00	\$ 18,413.50	\$ 20,034.18
Plus: Depreciation		20,000.00	20,000.00	20,000.00	20,000.00	20,000.00
Less: CAPEX	(100,000.00)	-	-	-	-	-
Less: Change in NWC	(5,000.00)	(250.00)	(262.50)	(275.63)	(289.41)	6,077.53
Project Free Cash Flow	\$ (105,000.00)	\$ 33,750.00	\$ 35,137.50	\$ 36,594.37	\$ 38,124.09	\$ 46,111.71

Problem 2

$$a. \quad NPV = -105,000 + \frac{33,750}{(1+11.24\%)^1} + \frac{35,137.50}{(1+11.24\%)^2} + \frac{36,594.37}{(1+11.24\%)^3} + \frac{38,124.09}{(1+11.24\%)^4} + \frac{46,111.71}{(1+11.24\%)^5} = 32,288.55 > 0$$

$$\bullet \quad NPV = 0 = -105,000 + \frac{33,750}{(1+IRR)^1} + \frac{35,137.50}{(1+IRR)^2} + \frac{36,594.37}{(1+IRR)^3} + \frac{38,124.09}{(1+IRR)^4} + \frac{46,111.71}{(1+IRR)^5}, IRR = 22.29\%$$

b. EVA

$$EVA_t^{\text{®}} = \frac{\text{Net Operating Profit after Taxes (NOPAT)}_t}{\text{Capital Charge}_t} \quad (7.2)$$

where

$$NOPAT_t = \left(\frac{\text{Earnings Before Interest and Taxes (EBIT)}_t}{\text{EBIT}_t} \right) \times \left(1 - \frac{\text{Corporate Tax Rate}}{\text{Tax Rate}} \right)$$

and

$$\text{Capital Charge}_t = \frac{\text{Invested Capital}_{t-1}}{\text{Capital}_{t-1}} \times \frac{\text{Weighted Average Cost of Capital (WACC)}}{\text{Capital (WACC)}}$$

Problem 2

- $Invested\ capital_{2016} = 105,000$; $Capital\ charge_{2016} = 105,000 \times 11.24\% = 11,802$; $EVA_{2016} = 14,000 - 11,802 = 2,198$
- $Invested\ capital_{2017} = 105,000 - 20,000 - (-200) = 85,250$; $Capital\ charge_{2017} = 85,250 \times 11.24\% = 9582.1$; $EVA_{2017} = 15,400 - 9582.1 = 5,817.9$
- $EVA_{2018} = 9506.4$; $EVA_{2019} = 13,266.91$; $EVA_{2020} = 17,103.06$
- $MVA = \frac{2,198}{(1+11.24\%)^1} + \frac{5,817.9}{(1+11.24\%)^2} + \frac{9,506.4}{(1+11.24\%)^3} + \frac{13,266.91}{(1+11.24\%)^4} + \frac{17,103.06}{(1+11.24\%)^5} = 32,288.55$

Problem 2

- c. If the project's economic profit in 2016 and 2017 were both negative, this could create the perception that the project is underperforming, especially if managers are evaluated based on short-term EVA results. To avoid rejecting value-creating projects, the firm could adopt performance evaluation tools that focus on long-term value, such as revised EVA or bonus bank.

- *Thank you for your time and attention!*